# Basic Inferential Data Analysis: The ToothGrowth Data

Simon Malone

02 November 2026

## Overview

The `ToothGrowth` dataset (R `datasets` package) records the effect of vitamin C on tooth growth in guinea pigs. Each of 60 animals received one of three dose levels of vitamin C (0.5, 1 or 2 mg/day) by one of two delivery methods: orange juice (`OJ`) or ascorbic acid (`VC`). The response, `len`, is the length of odontoblasts (cells responsible for tooth growth). This report explores the data, then uses confidence intervals and t-tests to compare tooth growth by delivery method (`supp`) and by `dose`.

## 1. Exploratory Data Analysis

The data structure, `summary()` output and the supplement x dose counts are shown in Appendix A.

## 2. Summary of the Data

The data are balanced: 10 animals in each of the 6 supplement x dose combinations. The table below gives the mean and standard deviation of tooth length for each combination.

Table 1: Table 1. Tooth length by supplement and dose.

| Supplement | Dose (mg/day) | n | Mean length | SD |
|---|---|---|---|---|
| OJ | 0.5 | 10 | 13.23 | 4.46 |
| VC | 0.5 | 10 | 7.98 | 2.75 |
| OJ | 1.0 | 10 | 22.70 | 3.91 |
| VC | 1.0 | 10 | 16.77 | 2.52 |
| OJ | 2.0 | 10 | 26.06 | 2.66 |
| VC | 2.0 | 10 | 26.14 | 4.80 |

Two features stand out:

1. **Tooth length rises steadily with dose** for both delivery methods.
2. **Orange juice appears more effective than ascorbic acid at 0.5 and 1 mg**, but at 2 mg the two methods look the same (both means about 26).

The formal tests below check whether these impressions are statistically supported.

## 3. Comparing Tooth Growth by Supplement and Dose

All comparisons use **Welch two-sample t-tests / 95% confidence intervals** (unpaired, unequal variances), since different animals are in each group. Each difference is reported as (first group - second group).

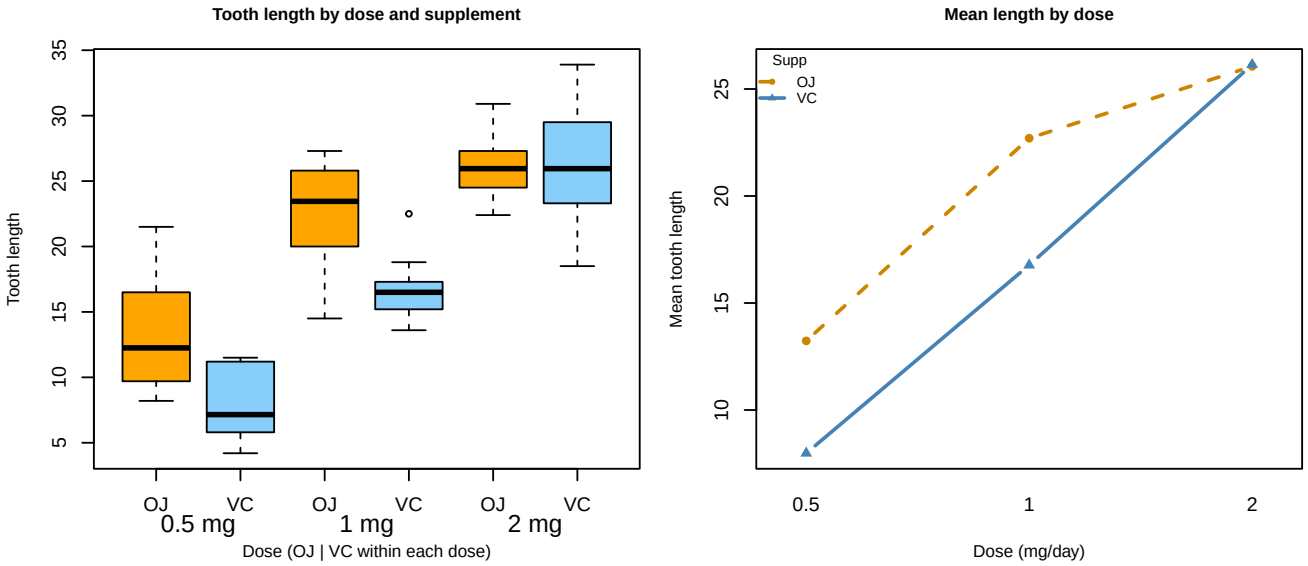



Figure 1: Figure 1. Tooth length by dose, split by delivery method (left) and mean tooth length for each delivery method across doses (right).

Table 2: Table 2. Welch t-tests and 95% confidence intervals (7 comparisons).

| Comparison | Mean diff. | 95% CI lower | 95% CI upper | t | df | p-value | Bonferroni p |
|---|---|---|---|---|---|---|---|
| OJ - VC (all doses) | 3.70 | -0.17 | 7.57 | 1.92 | 55.31 | 0.0606 | 0.424 |
| OJ - VC at 0.5 mg | 5.25 | 1.72 | 8.78 | 3.17 | 14.97 | 0.00636 | 0.0445 |
| OJ - VC at 1 mg | 5.93 | 2.80 | 9.06 | 4.03 | 15.36 | 0.00104 | 0.00727 |
| OJ - VC at 2 mg | -0.08 | -3.80 | 3.64 | -0.05 | 14.04 | 0.964 | 1 |
| Dose 1 - 0.5 mg (both supps) | 9.13 | 6.28 | 11.98 | 6.48 | 37.99 | 1.27e-07 | 8.88e-07 |
| Dose 2 - 1 mg (both supps) | 6.37 | 3.73 | 9.00 | 4.90 | 37.10 | 1.91e-05 | 0.000133 |
| Dose 2 - 0.5 mg (both supps) | 15.50 | 12.83 | 18.16 | 11.80 | 36.88 | 4.4e-14 | 3.08e-13 |

### Delivery method (supp)

- **Across all doses**, orange juice gives a mean 3.7 units more growth than ascorbic acid, but the 95% CI (-0.17 to 7.57) contains 0 and p = 0.061. At the 5% level there is **no significant overall difference**.
- **At 0.5 mg**, OJ is better by 5.25 (95% CI 1.72 to 8.78, p = 0.0064), and **at 1 mg** by 5.93 (95% CI 2.8 to 9.06, p = 0.001). Both intervals exclude 0, and both remain significant after Bonferroni correction for 7 tests.
- **At 2 mg**, the difference is -0.08 (95% CI -3.8 to 3.64, p = 0.96): there is no evidence of any difference.

The overall null result is therefore driven by the 2 mg group masking a real OJ advantage at the lower doses.

### Dose

- Going from 0.5 to 1 mg increases mean tooth length by 9.13 (95% CI 6.28 to 11.98), from 1 to 2 mg by 6.37 (95% CI 3.73 to 9), and from 0.5 to 2 mg by 15.5 (95% CI 12.83 to 18.16).
- All three intervals are well above 0 and all p-values are tiny (see Table 2), even after Bonferroni correction. **Higher doses of vitamin C produce longer teeth**, and this holds for each step up in dose.

## 4. Conclusions and Assumptions

### Conclusions

1. Vitamin C dose has a strong, statistically significant positive effect on tooth growth: each increase (0.5 to 1 to 2 mg/day) raises mean tooth length.
2. Delivery method matters at low and medium doses: orange juice produces significantly more growth than ascorbic acid at 0.5 mg and 1 mg.
3. At 2 mg/day the two delivery methods perform equally well, and over all doses combined the supplement difference is not significant at the 5% level.

### Assumptions needed

- **Independence / random assignment**: the 60 guinea pigs are independent, and animals were assigned to the dose and supplement groups at random (so that differences can be attributed to the treatments rather than to confounders such as litter or age). The documentation does not fully confirm this.
- **Unpaired groups**: each animal appears in exactly one group, so independent-group (not paired) tests are appropriate.
- **Approximate normality**: the t procedures assume tooth lengths within each group are roughly normal (or that the data are not too skewed). With only 10 animals per cell this cannot be checked well, and it is the main limitation; the box plots show no strong skew and only two mild outliers (one in VC at 1 mg, one in OJ at 2 mg).
- **Unequal variances**: Welch's test does not require equal variances, and group SDs do differ (e.g. VC at 2 mg has SD 4.8 versus 2.7 for OJ at 2 mg), so this choice is the safer one.
- **Dose treated as a category**: I compared the three dose levels as separate groups rather than assuming a linear dose-response.
- **Multiple testing**: I ran seven tests, so a few "significant" results could occur by chance; the Bonferroni-adjusted p-values show the main conclusions survive this correction. Conclusions are also limited to these doses and to guinea pigs.

## Appendix A: Supporting Output

Data structure, summary statistics and group counts referred to in Sections 1 and 2.

```
str(ToothGrowth)
```

```
## 'data.frame':    60 obs. of  3 variables:
##  $ len : num  4.2 11.5 7.3 5.8 6.4 10 11.2 11.2 5.2 7 ...
##  $ supp: Factor w/ 2 levels "OJ","VC": 2 2 2 2 2 2 2 2 2 2 ...
##  $ dose: num  0.5 0.5 0.5 0.5 0.5 0.5 0.5 0.5 0.5 0.5 ...
```

```
summary(ToothGrowth)
```

```
##       len        supp         dose
##  Min.   : 4.20   OJ:30   Min.   :0.500
##  1st Qu.:13.07   VC:30   1st Qu.:0.500
##  Median :19.25           Median :1.000
##  Mean   :18.81           Mean   :1.167
##  3rd Qu.:25.27           3rd Qu.:2.000
##  Max.   :33.90           Max.   :2.000
```

```
table(Supplement = tg$supp, Dose = tg$dose)
```

```
##           Dose
## Supplement 0.5  1  2
##         OJ  10 10 10
##         VC  10 10 10
```

## Appendix B: R Code

Complete code used to produce this report.

```
# echo = FALSE keeps the code out of the main report; all code is listed in the Appendix
knitr::opts_chunk$set(echo = FALSE, fig.align = "center", fig.width = 7, fig.height = 3.4,
        out.width = "100%")
data(ToothGrowth)
tg <- ToothGrowth
tg$dose_f <- factor(tg$dose)    # dose as a 3-level factor for grouping
str(ToothGrowth)
summary(ToothGrowth)
table(Supplement = tg$supp, Dose = tg$dose)
cell <- do.call(data.frame,
                aggregate(len ~ supp + dose, data = tg,
                        FUN = function(x) c(n = length(x), mean = mean(x), sd = sd(x))))
names(cell) <- c("Supplement", "Dose (mg/day)", "n", "Mean length", "SD")
knitr::kable(cell, digits = 2, caption = "Table 1. Tooth length by supplement and dose.")
par(mfrow = c(1, 2), cex = 0.66, mar = c(4, 4, 3, 1), cex.main = 0.9, cex.lab = 0.9)

boxplot(len ~ supp + dose, data = tg,
        col = rep(c("orange", "lightskyblue"), 3),
        names = rep(c("OJ", "VC"), 3), xlab = "Dose (OJ | VC within each dose)",
        ylab = "Tooth length", main = "Tooth length by dose and supplement")
mtext(c("0.5 mg", "1 mg", "2 mg"), side = 1, line = 1.9, at = c(1.5, 3.5, 5.5), cex = 0.8)
```

```r
with(tg, interaction.plot(dose_f, supp, len, type = "b", pch = c(16, 17),
                          col = c("orange3", "steelblue"), lwd = 2, legend = FALSE,
                          xlab = "Dose (mg/day)", ylab = "Mean tooth length",
                          main = "Mean length by dose"))
legend("topleft", title = "Supp", legend = levels(tg$supp), bty = "n",
       col = c("orange3", "steelblue"), pch = c(16, 17), lty = c(2, 1), lwd = 2,
       cex = 0.7)
par(mfrow = c(1, 1))
tt <- function(x, y) t.test(x, y, paired = FALSE, var.equal = FALSE, conf.level = 0.95)

row <- function(label, t) {
  data.frame(Comparison = label,
             Difference = unname(t$estimate[1] - t$estimate[2]),
             CI_lower   = t$conf.int[1],
             CI_upper   = t$conf.int[2],
             t          = unname(t$statistic),
             df         = unname(t$parameter),
             p          = t$p.value)
}

L <- function(s = NULL, d = NULL) {
       keep <- rep(TRUE, nrow(tg))
       if (!is.null(s)) keep <- keep & tg$supp == s
       if (!is.null(d)) keep <- keep & tg$dose %in% d
       tg$len[keep]
}

res <- rbind(
       row("OJ - VC (all doses)",  tt(L("OJ"), L("VC"))),
       row("OJ - VC at 0.5 mg",    tt(L("OJ", 0.5), L("VC", 0.5))),
       row("OJ - VC at 1 mg",      tt(L("OJ", 1),   L("VC", 1))),
       row("OJ - VC at 2 mg",      tt(L("OJ", 2),   L("VC", 2))),
       row("Dose 1 - 0.5 mg (both supps)", tt(L(d = 1), L(d = 0.5))),
       row("Dose 2 - 1 mg (both supps)",   tt(L(d = 2), L(d = 1))),
       row("Dose 2 - 0.5 mg (both supps)", tt(L(d = 2), L(d = 0.5)))
)

# Bonferroni adjustment across all 7 tests
res$p_bonferroni <- pmin(1, p.adjust(res$p, method = "bonferroni"))

out <- res
out$p            <- formatC(res$p,            format = "g", digits = 3)
out$p_bonferroni <- formatC(res$p_bonferroni, format = "g", digits = 3)
knitr::kable(out, digits = 2, row.names = FALSE,
             col.names = c("Comparison", "Mean diff.", "95% CI lower", "95% CI upper",
             "t", "df", "p-value", "Bonferroni p"),
             caption = "Table 2. Welch t-tests and 95% confidence intervals (7 comparisons).")
```